

SONiX MCU CRC Tool

User Guide

Version 1.0

SN8F5830

SN32F280/290

SN32F780/790

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AMENDENT HISTORY

Version	Date	Description
1.0	2020/06/24	First version is released.

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1 OVERVIEW

The purpose of this document is to make the users be familiar with SONiX CRC Calculator Tool and the settings of Keil MDK-ARM.

1.1 CRC Calculator Tool's features

- Support 8051 and M0 series MCU.
- In accordance with different IC series auto calculate ROM CRC checksum.
- Provide a convenient method for verifying CRC result.

2 Environment Setup

2.1 8051 Environment Setup

- (1). Please link to <http://www.sonix.com.tw/> to download SN-Link Driver for C51.
- (2). Install “SN-Link_Driver for C51_V2.00.37.exe” to default path (C:\Keil_v5\UV4)
 - Please check the driver version is V2.00.37 or above.

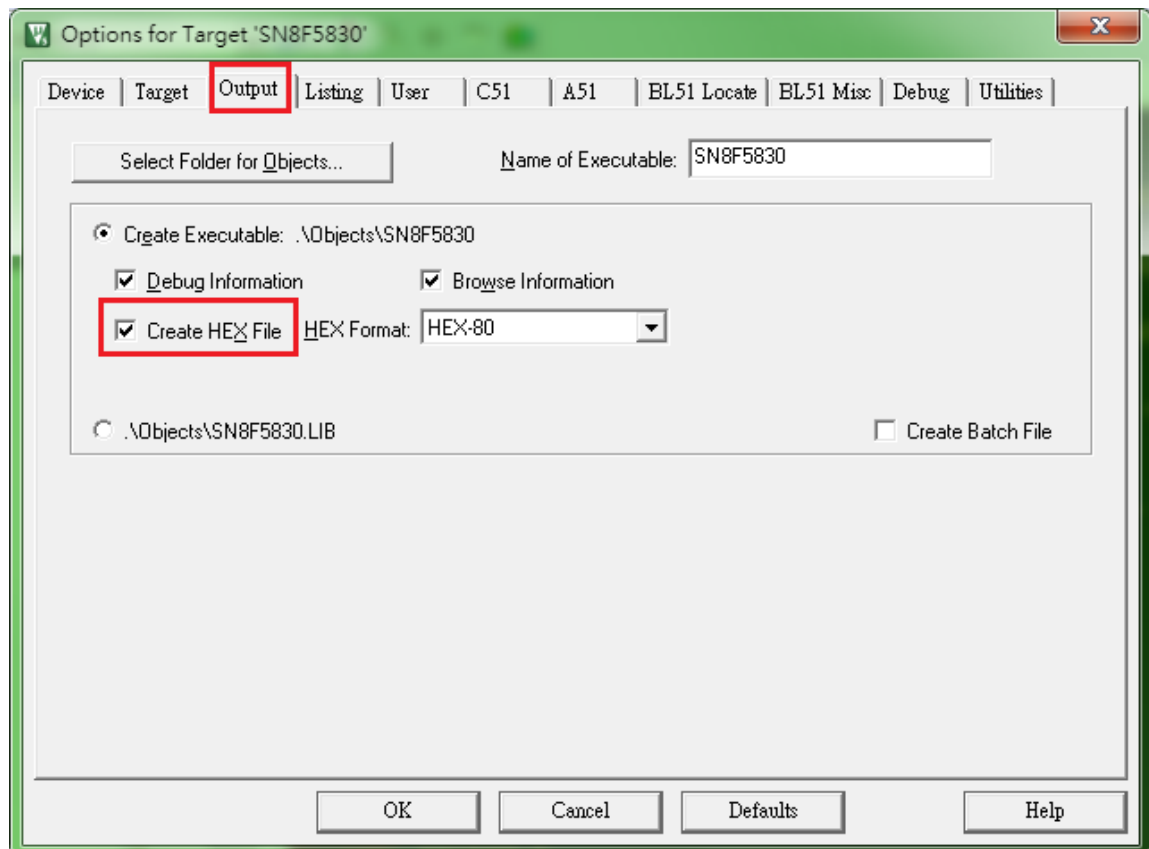
2.2 M0 Environment Setup

- (1). Please link to <http://www.sonix.com.tw/> to download SN-Link Driver for M0.
- (2). Install “SN-Link_Driver for Keil Cortex M0_V2.00.319.exe” to default path (C:\Keil_v5\UV4)
 - Please check the driver version is V2.00.319 or above.

3 Calculate CRC Result

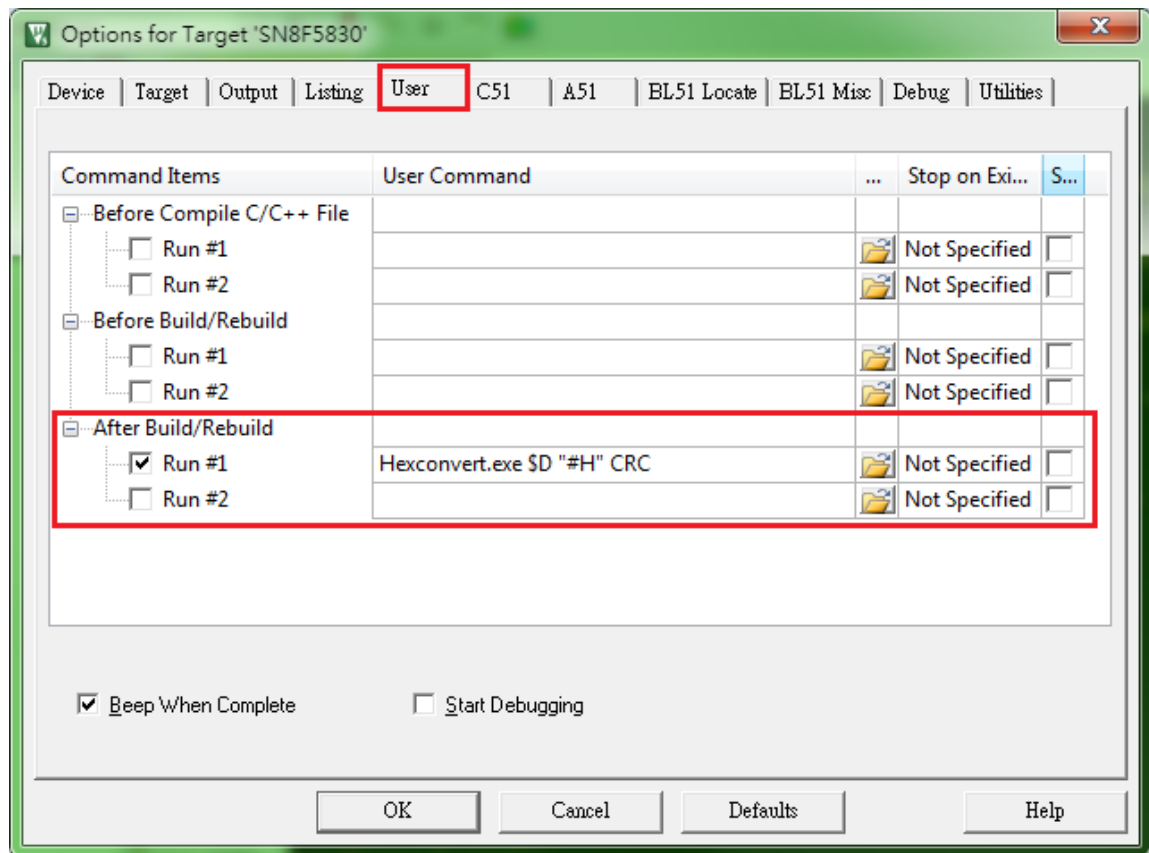
3.1 8051 Environment Setup (Taking SN8F5835 for example)

- Press “Output”, and check “Create HEX File.”



➤ Press “User”, check “Run#1:”, and then fill in the command below for each MCU type.

1. SN8F5835 → HexConvert.exe \$D "#H" CRC.



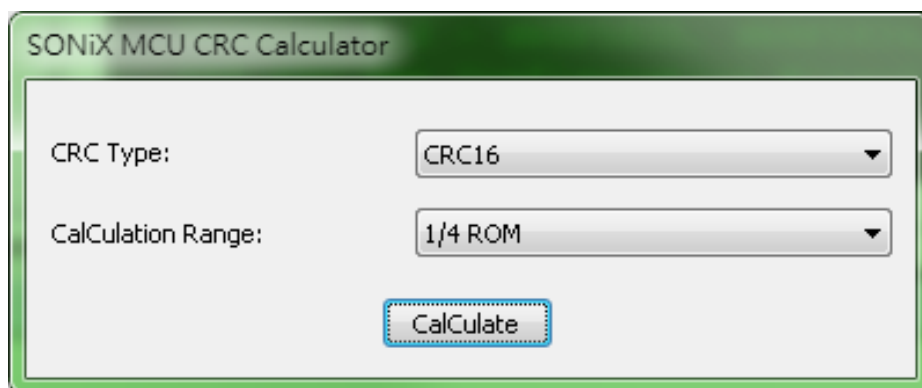
- **Return to main page to start coding.**
- **After coding, click the “Rebuild” button as below, Keil MDK will start to compile.**



Build(F7)

- **If compile successfully, the SONiX MCU CRC Calculator APP will run.**

1. Please select correct CRC Type.
 - CRC16
 - CRC16-CCITT
2. Please select correct Calculation Range.
 - 1/4 ROM
 - 1/2 ROM
 - Avoid the last 8 page
 - Avoid the last 1 page + 2 Byte

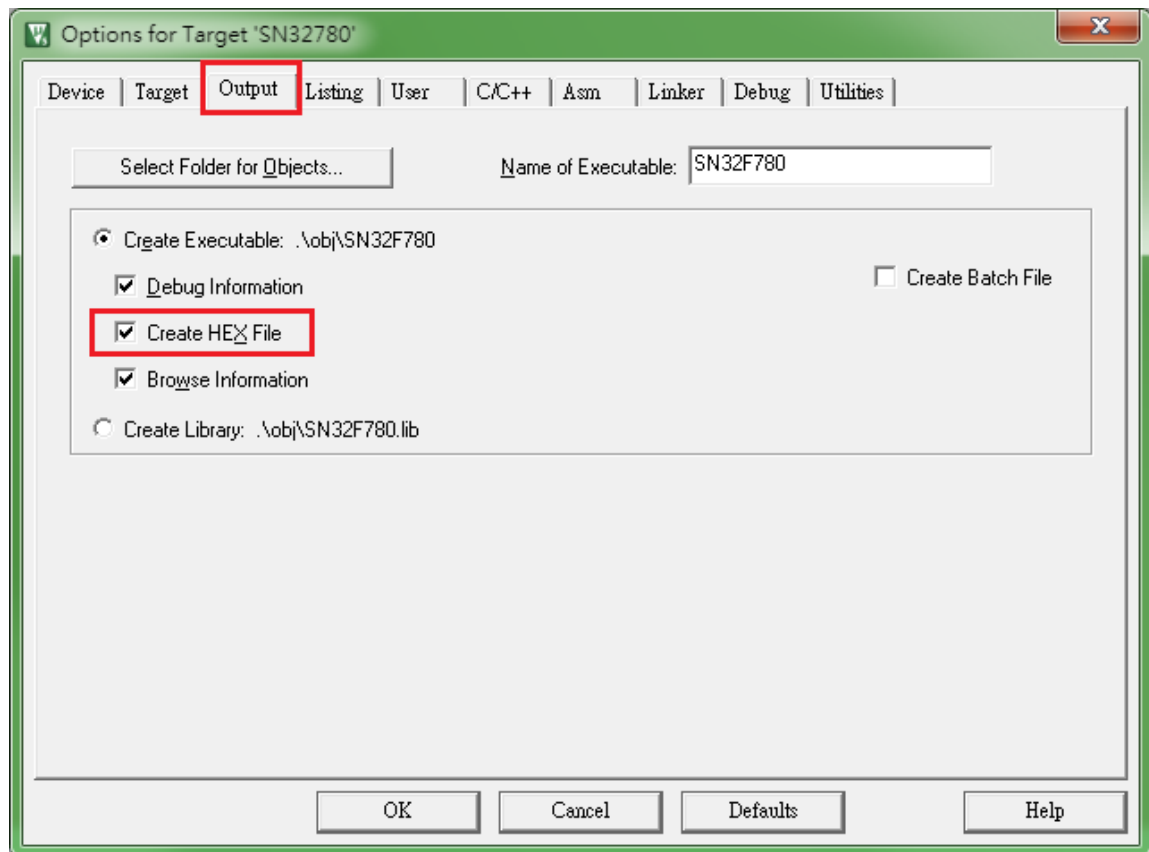


- **APP will calculate CRC result and shown in the message box after press “Calculate”.**

```
CRC_Type = CRC16
CRC_CalculationRange = 1/4 ROM
CRC_Checksum = 0xB7D7
```

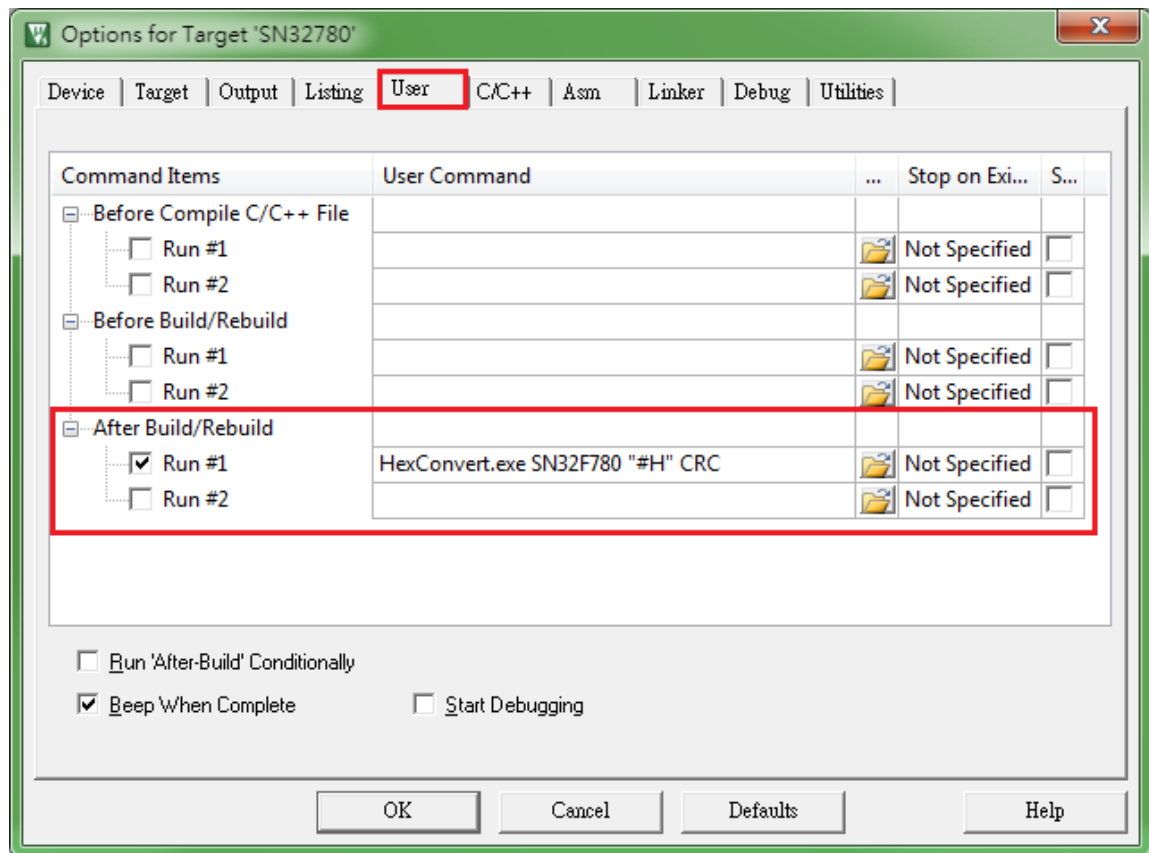
3.2 M0 Environment Setup (Taking SN32F780 for example)

- Press “Output”, and check “Create HEX File.”

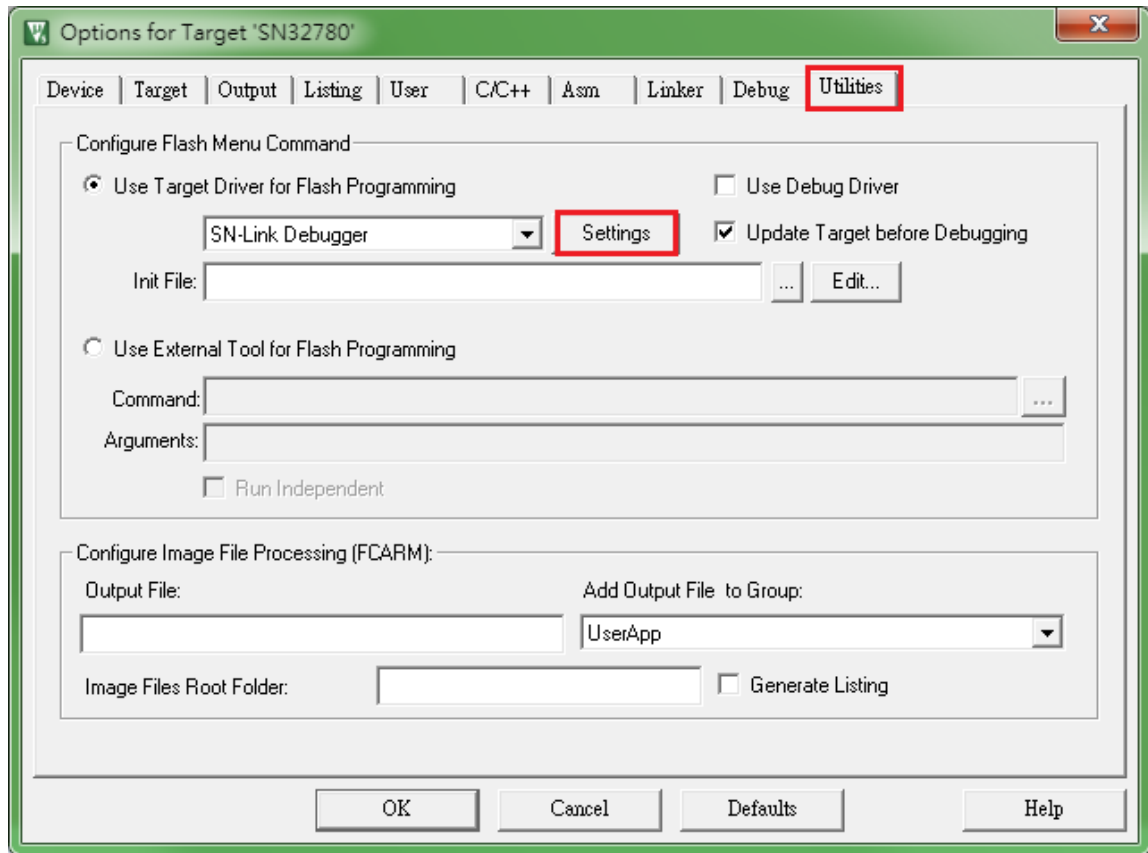


➤ Press “User”, check “Run#1:”, and then fill in the command below for each MCU type.

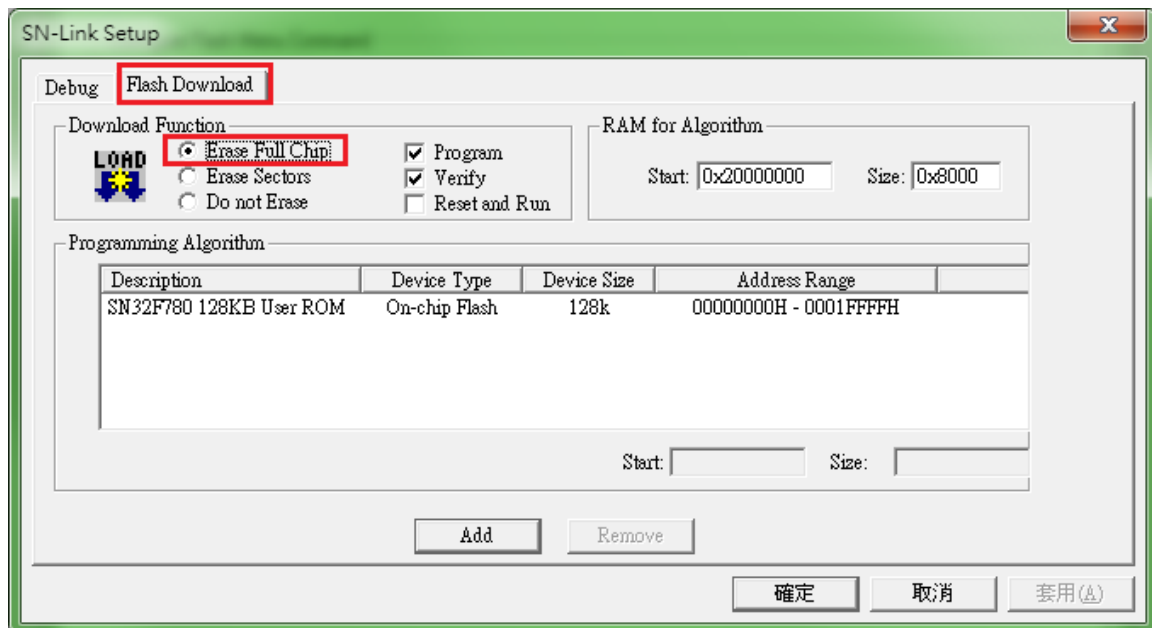
1. SN32F780 → HexConvert.exe SN32F780 "#H" CRC.



- Press “Utility”, select the used ICE, and then press “Settings”.



- Configure and choose the correct Programming algorithm for MCU in use as below.



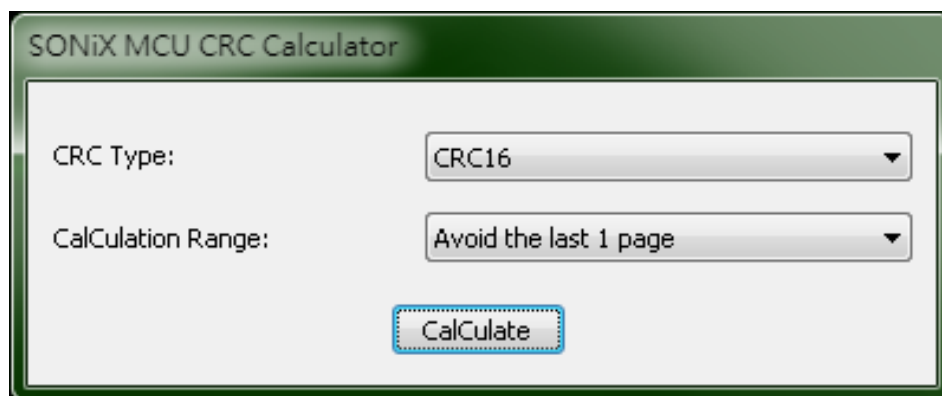
- **Return to main page to start coding.**
- **After coding, click the “Rebuild” button as below, Keil MDK will start to compile.**



Build(F7)

- **If compile successfully, the SONiX MCU CRC Calculator APP will run.**

1. Please select correct CRC Type.
 - CRC16
 - CRC16-CCITT
 - CRC32
2. Please select correct Calculation Range.
 - Avoid the last 1 Page



- **APP will calculate CRC result and shown in the message box after press “Calculate”.**

```
CRC_Type = CRC16
CRC_CalculationRange = Avoid the last 1 page
CRC_Checksum = 0x61B4
```

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